mixmaster singularity. A singularity near which tidal gravity oscillates chaotically with time, but does not necessarily vary in space. See also *BKL singularity.*

molecule. An entity made of several atoms that share their electron clouds with each other. Water is a molecule made in this way from two hydrogen atoms and one oxygen.

mouth. An entrance to a wormhole. There is a mouth at each of the two ends of the wormhole.

naked singularity. A singularity that is not inside a black hole (not surrounded by a black-hole horizon), and that therefore can be seen and studied by someone outside it. See *cosmic censorship conjecture.*

National Science Foundation (NSF). The agency of the United States government charged with the support of basic scientific research.

natural philosopher. A phrase widely used in the seventeenth, eighteenth, and nineteenth centuries to describe what we now call a scientist.

nebula. A cloud of brightly shining gas in interstellar space. Before the 1930s, galaxies were widely, but not universally, mistaken for nebulas.

neutrino. A very light particle that resembles the photon, except that it interacts hardly at all with matter. Neutrinos produced in the Sun's center, for example, fly out through the Sun's surrounding matter without being absorbed or scattered hardly at all.

neutron. A subatomic particle. Neutrons and protons, held together by the nuclear force, make up the nuclei of atoms.

neutron core. Oppenheimer's name for a neutron star. Also a neutron star at the center of a normal star.

neutron star. A star, about as massive as the Sun but only 50 to 1000 kilometers in circumference, and made from neutrons packed tightly together by the force of gravity.

new quantum mechanics. The final version of the laws of quantum mechanics, formulated in 1926.

Newtonian laws of physics. The laws of physics, built on Newton's conception of space and time as absolute, which were the centerpiece of nineteenth-century thinking about the Universe.

Newton's law of gravity. See *inverse square law of gravity.*

no-hair conjecture. The conjecture in the 1960s and 1970s (which was proved to be true in the 1970s and 1980s) that all the properties of a